Agent Team Debugger study — facilitator packet

Participant information sheet and consent form

Study: Interactive debugging of multi-agent AI systems (CS702 term project, Singapore Management University) **Researchers:** Nguyen Nhat Minh (nm.nguyen.2026@phdcs.smu.edu.sg), Do Duc Anh (doducanh.2026@phdcs.smu.edu.sg) **Course instructor:** Prof. Kotaro Hara

What the study is about

We re-implemented a research tool for debugging teams of AI agents (large-language-model agents that exchange messages to solve a task) and want to learn how people use it to find and fix errors in recorded agent runs.

What you will do

- Part 1 (about 50 minutes): after a short demo, you will look at three recorded runs of an agent team that each ended with a wrong answer, using three versions of the tool (15 minutes each), and tell us where you think the run went wrong. You will then say which version you preferred and why.
- Part 2 (about 60 minutes): after a short demo, you will try to steer one failing run to the correct answer by editing the agents' messages (30 minutes), thinking aloud as you work, then answer a short questionnaire and a few interview questions.

You will be assigned to one part only. No programming is required; some familiarity with LLM chat tools is assumed.

Data we collect

- A screen recording of the tool during the task and an audio recording of what you say while thinking aloud.
- The tool's interaction log (clicks, edits, messages flagged) stored under a participant number, not your name.
- Your questionnaire answers.

No personal data beyond your name on this form is kept; the form is stored separately from the data. Recordings are used only for analysis by the two researchers, are not shared, and are deleted at the end of the course (December 2026). Quotes may appear in the course report, attributed to a participant number only. The agent runs you will see involve no personal data; the "web pages" the agents read are a fictional corpus and public Wikipedia articles.

Voluntary participation

Participation is voluntary. You may stop at any time without giving a reason and ask for your data to be deleted. There is no payment and no effect on any course grade.

Consent

I have read the information above, had the chance to ask questions, and agree to take part. I agree to screen and audio recording as described.

Name: _____ Signature: _____ Date: _____

Participant number (filled by the researcher): _____

Task sheets

Participant-facing sheets are the boxed parts; everything under **Facilitator** stays with the facilitator. Each sheet mirrors what the paper gave its participants: the task, the expected output, the incorrect output.

The three Part 1 recordings are natural failures of the same Claude Haiku team, 224–233 runtime messages each (119–134 visible cards). Tasks A and B are two runs of the same corpus question; Task C is a live-Wikipedia question. Because A and B share a question, the schedule below never gives a participant A and B in consecutive positions with the same condition, and the analysis reports condition × task cells and position effects.

Assignment schedule (Part 1, n = 6)

Every condition meets every task twice; every condition appears in every position twice; A and B are separated by C for four of six participants.

Participant	Position 1	Position 2	Position 3
P01	reduced / A	full / C	e1 / B
P02	full / B	e1 / A	reduced / C
P03	e1 / C	reduced / B	full / A
P04	reduced / B	e1 / C	full / A
P05	full / C	reduced / A	e1 / B
P06	e1 / A	full / B	reduced / C

Enter the same strings in `data/participants.csv` (`condition_order`, `task_order`), e.g. `reduced,full,e1` and `A,C,B` for P01. URLs: `?p=P01&cond=reduced&trial=A`, then `?p=P01&cond=full&trial=C`, then `?p=P01&cond=e1&trial=B`. Opening the URL is the trial's t = 0 in the log; start the stopwatch at the same moment.

Task A

The agents were asked: Using the Lumen Valley Observatory's own figures, compute the number of visitors admitted in its financial year 2024/25 divided by the number of instruments it operated on the last day of that financial year. Reply with the result rounded to one decimal place.

Correct answer: 6651.7

What the agents produced: after 228 messages the orchestrator wrote that no number could be produced and that "no source was actually checked".

Your job (15 minutes): find where this run went wrong. Flag every message you consider an error (□ on the card, short note) and tell us, thinking aloud, what the error is. You do not need to fix anything.

Facilitator: recording `t13_fy_ratio_FAIL`. Ground truth: `ground_truth/t13_fy_ratio_FAIL.json` (primary errors t=23 searcher reports the monthly tables as unusable; t=25 orchestrator forbids calendar-year figures). The answer comes from summing the twelve monthly figures in the two Annual Visitor Reports and dividing by the six instruments on the Facilities page.

Task B

The agents were asked: Using the Lumen Valley Observatory's own figures, compute the number of visitors admitted in its financial year 2024/25 divided by the number of instruments it operated on the last day of that financial year. Reply with the result rounded to one decimal place.

Correct answer: 6651.7

What the agents produced: after 233 messages the orchestrator wrote that the figure could not be determined from the observatory's own data. This is a different run of the team from any you may have seen: read it on its own.

Your job (15 minutes): find where this run went wrong. Flag every message you consider an error (□ on the card, short note) and tell us, thinking aloud, what the error is. You do not need to fix anything.

Facilitator: recording `t13_fy_ratio_FAIL_2`. Ground truth: `ground_truth/t13_fy_ratio_FAIL_2.json` (primary errors t=36 orchestrator rejects a correct derivation; t=47 orchestrator tells the Coder not to present 6651.7).

Task C

The agents were asked: In the English Wikipedia article 'List of countries and dependencies by area', what rank is Mongolia given? Reply with just the number.

Correct answer: 18

What the agents produced: after 224 messages the orchestrator answered "my best estimate from memory is 19".

Your job (15 minutes): find where this run went wrong. Flag every message you consider an error (on the card, short note) and tell us, thinking aloud, what the error is. You do not need to fix anything.

Facilitator: recording `wt2_mongolia_rank_FAIL`. Ground truth: `ground_truth/wt2_mongolia_rank_FAIL.json`. Do not tell the participant that the fetch tool returns article text without tables, or that the Mongolia article states "18th-largest"; both are things to discover from the messages (t=15 shows table-less text, t=34 shows the Mongolia article). If asked what the tools can do, answer only "what you can see in their results".

Part 2 · Steering task

The agents were asked: Using the Lumen Valley Observatory's own figures, compute the number of visitors admitted in its financial year 2024/25 divided by the number of instruments it operated on the last day of that financial year. Reply with the result rounded to one decimal place.

Correct answer: 6651.7

What the agents produced: no answer after 228 messages.

Summary of errors found in Part 1 (given to you, as in the paper's protocol): the searcher reported the monthly visitor tables as unusable; the orchestrator then forbade using calendar-year figures, chased annual accounts that do not exist, and re-planned without instructing any agent.

Your job (30 minutes): steer the team so that *the agents themselves* answer 6651.7. You may edit any message and revert to it, send new messages, step or run the queue, and use the suspect panel. Each run continues live (the agents call a real model), so a branch takes one to several minutes to finish. Think aloud: say what you expect before you press Run. Typing the answer into a message yourself does not count.

Facilitator: recording `t13_fy_ratio_FAIL`, URL `?p=P1x&cond=e1&trial=S`. Success = the orchestrator's final message in a branch states 6651.7 (hedged wording still counts, as in our harness grader) and the participant did not write the figure into any message. Hard stop at 30:00: a branch still running at 30:00 is recorded as *unfinished* (let it run while the questionnaire starts and note its eventual outcome separately).

Questionnaires and interview guides

Items marked (paper) are taken verbatim from Appendix B of Epperson et al. (CHI 2025) so that means can be compared.

Pre-study (all participants)

1. How much experience do you have developing AI agents? (5-point: 1 none … 5 extensive) (paper)
2. Do you have prior experience using AutoGen or AgentChat? (Yes / No) (paper, "autogen or agnext")

3. Do you have prior experience with the GAIA benchmark? (Yes / No) (paper)
4. How often do you use LLM chat tools? (daily / weekly / monthly / rarely)
5. Programme and year (e.g. MSc CS, year 1)

Part 1 — after each condition (15-minute task)

Recorded by the facilitator from the participant's spoken summary; the participant also flags messages in the tool.

1. Where did the run go wrong? List the messages (by timestamp) and say what the error was. (open)
2. How confident are you that you found the main cause? (1 not at all … 5 very confident)
3. How much of the run did you feel you had to read? (1 almost none … 5 all of it)

Part 1 — after all three conditions

1. Which version did you prefer for finding errors: reduced, full, or full with suspects? Why? (paper: preference between full and reduced)
2. Rank the three versions from most to least helpful.
3. (full with suspects only) Did the suspects help, mislead, or make no difference? Give an example.

Part 2 — post-task survey (paper, Appendix B; 5-point Likert, 1 strongly disagree … 5 strongly agree)

4. I found this system helpful.
5. I found this system easy to use.
6. I would like to use this system again in the future.
7. Rate how much you found this feature helpful: Sending new messages.
8. Rate how much you found this feature helpful: Backtracking to previously sent messages and editing.
9. Rate how much you found this feature helpful: Conversation overview visualization.
10. Rate how much you found this feature helpful: Error context panel (suspects and context). *(our addition)*

Paper means for comparison: helpful 4.4, easy to use 4.0, use again 4.6, send 4.1, backtrack-and-edit 4.9, overview 4.1.

Part 2 — exit interview (paper, Appendix B)

1. (If did not steer to right answer) Given more time, do you think you would have been able to steer the models to output the correct answer?
2. Can you describe your approach to the task, how did you try to intervene?
3. After intervention, how has your opinion about which issues are the most actionable changed?
4. How did you feel that having the ability to edit messages and configuration influenced your understanding of the agents' errors?
5. What other ways would you have liked to steer the models?
6. *(ours)* Did the suspect list change where you made your first edit? Did you ever disagree with it?

Edit coding scheme (Part 2, applied by both researchers independently, disagreements resolved by discussion)

Code	Definition (paper, Section 6.4)	Example
add	adds more specific instructions or makes an instruction more concrete	"fetch both reports completely, use the offset parameter"
simplify	removes text, narrows the instruction to a sub-task	drops the second half of a compound instruction
modify	changes the goal or the plan itself	"use code to sum the monthly table instead of searching for a total"

Also recorded per edit: timestamp edited, sender of the edited message, whether it was a judge suspect, session outcome (correct / incorrect / unfinished at 30 min). A mixed edit takes the code of its dominant change and is marked mixed; sends and queue edits are not coded; plain resets are counted separately.

Interview coding

Open-response answers are transcribed and coded inductively by both authors: first pass independently (one label per statement, labels proposed freely), then a merged codebook, then a second pass with the merged codebook; disagreements are resolved by discussion and counted. Themes reported in the paper (counterfactual testing, "did my edit work", over-complex instructions) are used as seed codes so that overlap and differences with the paper can be stated.

Facilitator script

Two facilitators per session where possible: one runs the session, one takes notes (timestamps of spoken error localisations, quotes). Record the screen and audio with QuickTime or OBS from the start of the demo.

Setup (5 min before the participant arrives)

1. `cd term_project/prototype && ./run.sh`; confirm `http://127.0.0.1:8000` loads.
2. Look up the participant row in `data/participants.csv` (id, part, condition order, task order).
3. Open the first condition URL, e.g. `http://127.0.0.1:8000/?p=P03&cond=reduced`, and **Load recording…** the first task's recording. Keep a second tab on the plain URL for the demo.
4. Print the task sheets for the assigned tasks and one consent form.

Welcome (5 min)

"Thanks for coming. We are evaluating a tool for debugging teams of AI agents. Please read and sign this form; you can stop at any time." Collect the form. Ask the pre-study questions (questionnaires.md) and note the answers in `data/participants.csv`.

Demo (6 min, same words and the same features for every participant)

The demo always shows the **full set of features in the `e1` condition**, whatever the participant's first condition, so that training is equal across orders. Use recording `t1_telescopes` (a short, passing run) at `?p=<id>&cond=e1&trial=demo`:

- "An agent team is an orchestrator that plans, and three workers: a Coder, an Executor that runs code, and a WebSearcher with search and fetch tools. They talk by messages. Each card in the middle is one message: who sent it, to whom, its type and its content. Tool calls and results are messages too."
- "The flag button on a card records that you think this message is an error; type a short note."
- "The right-hand overview shows the whole run: one square per message, coloured by type, sender or recipient. Hover previews, click jumps. Each column is one session; editing a message forks a new column."
- "Edit & revert rewinds every agent to just before that message, applies your edit and continues in a new session. Reset here does the same without changes. The queue on the left shows what the agents will process next: Run, Step one message, or Drop it. You can also send your own message to the group or to one agent."
- "The error context panel lists up to five suspect messages ranked by a judge model, with a reason. Click one to see who said what to that agent, the tool calls before it, warnings, and a suggested edit. Edit this message & revert opens the editor there."
- "In some of your tasks only part of these features will be available; the sheet tells you which."
- Let the participant try each control once (flag, overview click, Edit & revert then Cancel, open a suspect). Answer questions about the interface, not about the tasks.

Part 1 (3 × 15 min + 5 min questions)

Use the schedule in `task_sheets.md`. For each position:

1. Open the trial URL with the participant id, condition and task letter, e.g. `?p=P01&cond=reduced&trial=A`. The server logs `trial_start`; **start the stopwatch at the same moment**. Load the task's recording, hand over the sheet.
2. "You have 15 minutes. Find where this run went wrong. Flag the messages you consider errors and tell me, as you go, what the error is. Think aloud."
3. Note-taker: write down every spoken localisation with the stopwatch time and the message timestamp if stated; the flags are in the log already. Do not help; if asked whether something is an error, say "tell me what you think and why".
4. At 15:00 stop. Ask the three post-condition questions; write answers in `data/part1_localisation.csv`: one row per distinct error the participant *named but did not flag* (with the matched ground-truth id or `none`), plus one row per trial carrying the confidence and read-amount answers (ground_truth_id empty). Flagged messages are scored from the log automatically.
5. Next position: new URL (new `trial_start`), new recording, new sheet. Never reuse a tab from an earlier trial.

Matching rule for spoken localisations: a statement matches a ground-truth error when it names the same message (timestamp) or the same mistake in the same agent's turn (e.g. "the searcher said the tables were useless" matches E1 of Task A even without a timestamp); a statement that names a consequence only ("it kept re-planning") matches the stall-loop error; a statement can match at most one error, the earliest it fits; repeated mentions count once.

After the third position ask the preference questions; record in `data/participants.csv` (`preference`, `ranking`).

Part 2 (60 min)

1. Open `?p=P1x&cond=e1&trial=S`, load `t13_fy_ratio_FAIL`, hand over the Part 2 task sheet (with the error summary). Stopwatch starts when the URL opens.

2. "You have 30 minutes to steer the team so that the agents themselves answer 6651.7. Use anything in the tool. Say what you expect before each Run. A branch takes one to several minutes; you can keep reading while it runs."
3. Note-taker: for every edit, sent message or reset write the stopwatch time and a one-line description, and whether the participant opened the suspect panel first. The server log stores the full old and new text of every edit, queue edit and sent message, so coding can be done from the log.
4. Interruptions: if a model-error banner appears, say "that message was lost; use Reset here on it and Run again"; the time keeps running. If the backend crashes, restart it, reload the recording, and record the trial as *interrupted* in `data/ratings.csv` notes; do not restart the clock.
5. Hard stop at 30:00. Outcome: *correct* if a branch's final orchestrator message states 6651.7 and the participant never typed the figure into a message; *incorrect* if the branch finished with another answer; *unfinished* otherwise (a branch still running at 30:00 may complete while the questionnaire starts; note its eventual result in the notes column, it does not change the outcome).
6. Before loading anything else, **Save recording** as `P1x_part2` so the participant's branches are kept.
7. Questionnaire (items 4–10), then the exit interview (6 questions). Record audio; transcribe the answers to `data/ratings.csv` and the notes file.

Edit coding (both authors independently, from the log): code each *message edit* as add / simplify / modify using the scheme in `questionnaires.md`; a mixed edit takes the code of its dominant change (most characters changed), noted as mixed; sends and queue edits are logged as their own kinds and not coded; plain resets are counted separately. Keep both coders' labels in `data/part2_edits.csv` (add a column per coder) and report the number of disagreements before adjudication.

Closing

Thank the participant. Save the screen recording as `recordings_study/P<id>.mov` (not in git). Check that `logs/P<id>.jsonl` exists and ends with the last action. Fill the data CSVs the same day.